CSE590 · MIDTERM PROGRESS

When and why does ApET work?

Understanding reconstruction-based visual token compression in LLaVA-1.5-7B

APET IN ONE LINE

No attention scores for selection: keep hard-to-reconstruct features; merge discarded states into similar retained states.

SIMPLE, AND STRONG

At an average budget of 64 visual tokens, the paper reports 95.2% relative performance and 1.46× total inference speedup.

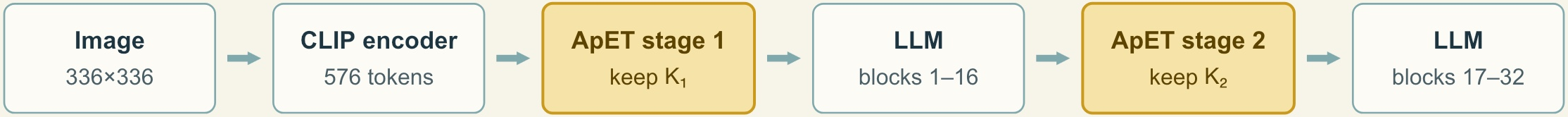
OUR QUESTION

How can a linear, geometric criterion work inside a nonlinear VLM, and when does it fail?

Peng Chen · SBU ID 117702477 · October 2026

Published results: Ma et al., CVPR 2026, Tables 1 and 7. Speed measured on POPE / single A100; not our Kaggle timing.

Method and three research questions



Setup: LLaVA-1.5-7B · native ApET code · Kaggle 2×T4, locked environment · GQA 800, MMStar 1,500, TextVQA 100 · avg. 64 or 192 tokens

RQ1 · Does E track answer sensitivity?

30 selected questions. Compare E ranking with an IH-inspired retain-one response: first reference-token probability relative to all visual states removed.

RQ2 · Selection and merging

Component pilots and broad native-versus-random comparisons are complete. Attribution remains open.

Next: controlled component tests on outcome-unselected cohorts.

RQ3 · An evidence-led improvement?

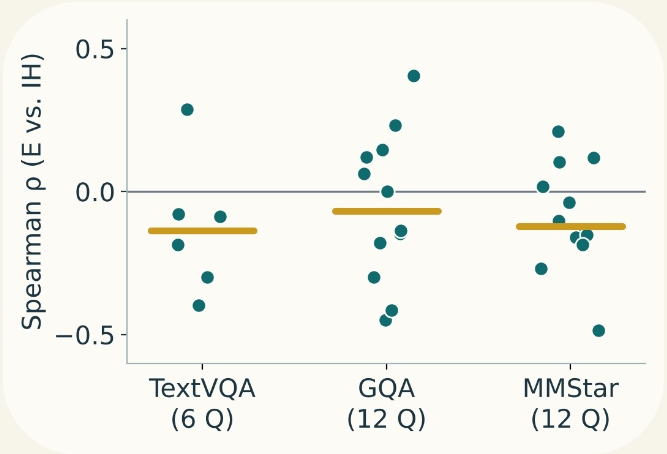
Conditional on the analysis. No validated improvement or adaptive policy yet.

E: reconstruction error from 10 FPS basis features. Two stages: 96→32 (avg. 64) or 288→96 (avg. 192). · 2 / 5

From reconstruction error to answer preservation

① E and answer sensitivity differ

IH-inspired retain-one test: first reference-token probability relative to removing all visual states. 30 selected questions.



Median correlations: −0.14 to −0.07.
Interpretation depends on the intervention.

② Merging has uneven roles

Earlier input-only288 panel: 24 images. Basis slots excluded from merge targets.

Share of incoming merges

Lowest-E retained quarter

66.1%

4,569 of 6,912 dropped sources

Highest-E retained quarter

1.8%

Protecting the top 28 rerouted only 21 / 6,912 sources.

③ Performance hides details

Native advantage varies by task and budget; similar scores hide answer turnover.

TextVQA100 · native − random

64 avg: +20.9 pts

192 avg: +9.4 pts

GQA800 · average budget 192

34 lost · 26 gained

Accuracy 69.0% → 68.0%

Smaller gap on GQA;
no clear gap on MMStar.

① Selected30; IH-inspired. ② Input288 / 24 images. ③ Staged cohorts; random = 3-seed mean. Sources: saved RQ1/RQ2 records. · 3 / 5

Roadblocks and fixes

A stable pipeline; remaining scientific limits are explicit

Roadblock	Problem	Action	Status
Integration	Kaggle package drift	Locked native environment	Resolved
Run-to-run variation	Fresh encodes vary; saved-feature replays agree.	Differences appear after merging; cause unconfirmed. Keep first captures; repeat critical cases.	Managed; cause open
Selection bias	Diagnostic cases include known native failures.	Use for case analysis; compare methods on outcome-unselected cohorts.	Limited scope
Planning risk	Component attribution remains open.	One controlled comparison; negative results also count.	Open

FP16 accumulation is a candidate explanation, not a confirmed root cause. Fixed-feature replay and fresh-run repeatability are different checks. · 4 / 5

Next steps follow the three findings

01 · Separate the components

Which operation helps?

E vs. random: basis + merge rule fixed.
Merge vs. drop: selection fixed.
Use an outcome-unselected cohort.

02 · Roles of retained tokens

Who collects information?

Check loads and feature changes across stages and budgets. Compare no, half-strength and native merging.

03 · Test reconstruction

Is reconstruction safe?

Replace a few features with linear reconstructions. Keep sequence length; control the change size.

SUCCESS CRITERION · RQ3 REMAINS CONDITIONAL

One controlled comparison on ≥100 outcome-unselected questions: matched budgets, 3 random seeds and paired intervals. Negative results count; a small improvement is optional.